

PIG LAUNCHER & RECEIVERS

PLR-QC Series

Secure Closure. Reliable Pigging. Assured Flow.



QUICK OPENING & CLOSURE

LINE*S*IG PLR-QC Series

Pig Launchers and Receivers provide reliable, safe and efficient solutions for pipeline cleaning, batching, gauging and intelligent inspection operations. Engineered for demanding oil, gas, petrochemical, water and process applications, each unit can be customized for pipeline size, pressure rating and service conditions, with quick-opening closure, pig signalling, skid-mounted options and complete inspection documentation.

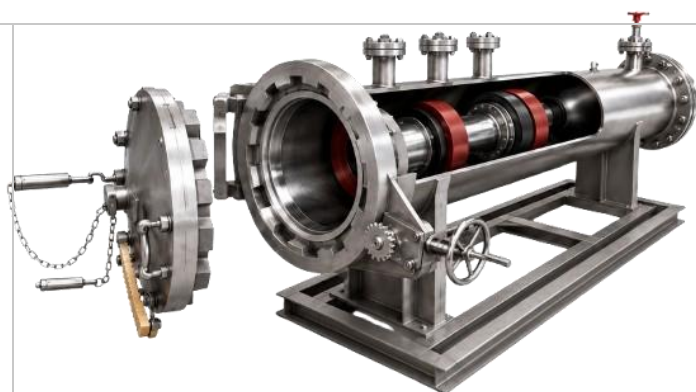
1. Product Overview

PLR-QC Series Pig Launcher & Receivers are designed for safe and repeatable insertion and recovery of pipeline pigs in oil, gas, petrochemical, water and process-pipeline service.

Units may accommodate cleaning pigs, batching pigs, spheres, gauging pigs and intelligent inspection pigs, subject to the approved design and operating envelope.

Standard supply can include a quick opening closure, pig signaler, vent and drain nozzles, supports, lifting lugs, hydrostatic testing, protective coating and manufacturing documentation.

Horizontal, vertical, permanent, modular and fully skid-mounted configurations can be engineered to suit the pipeline, site layout and maintenance philosophy.



Additional Product Pictures

Representative **PLR-QC Series** visuals for quick-opening closure type pig launcher and pig receiver assemblies are shown below.



Figure 1. Quick-opening closure pig launcher – closed front view.



Figure 2. Quick-opening closure pig receiver – skid-mounted product view.

2. Key Features and Product Benefits

Key Features

- Launcher and receiver configurations.
- Quick opening closure options.
- Pig signaller / indicator provision.
- Vent, drain, kicker and instrument nozzles.
- Field-supported or skid-mounted execution.
- Hydrostatic test before release.
- Project-specific coating and preservation.

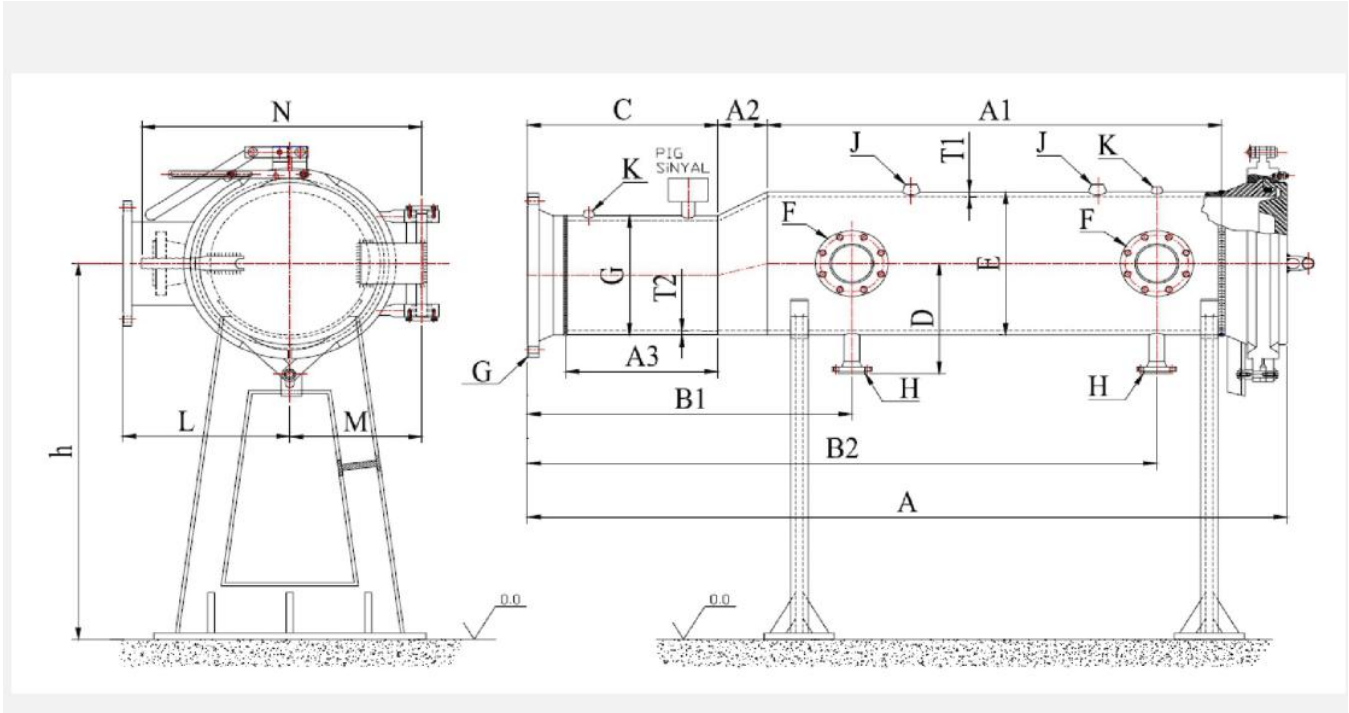
Product Benefits

- Supports efficient pipeline cleaning and inspection.
- Reduces field fabrication and installation work.
- Enables safer and faster pig handling.
- Supports pipeline integrity and product-quality programs.
- Customizable for line size, rating and service.
- Documentation package available for project records.

3. Typical Manufacturing Materials

Component	Typical Material Options
Main barrel / trunk	API 5L / ASTM A516 / project-specified carbon or alloy material
Nozzles and flanges	ASTM A105 / A350 LF2 / A182 F304 / F316 or equivalent
Quick opening closure	Forged or fabricated closure to approved design
Pig signaller and fittings	Carbon steel or stainless steel to suit service
Supports / skid	Structural carbon steel with specified coating
Seals / soft parts	NBR, FKM/Viton, PTFE, PUR or project-specified compound

4. Technical Drawing



Typical horizontal arrangement. Final nozzle orientation, support spacing and closure details are project-specific.

5. Weight & Dimensions

Weight & Dimensions

SIZE	A	B1	B2	C	D	E	T2	F	G	T1	H	J	K	L	M	N
3	41-13/16	18-1/4	33-1/8	10-3/4	3-7/8	4-1/2	0,144	3/4"	3"	0,219	2" Threaded	1" NPT	1/2" NPT	5-13/16	N/A	N/A
4	44-1/4	19-7/8	33-9/16	10-3/4	4-7/8	6-5/8	0,176	1"	4"	0,281	2" Threaded	1" NPT	1/2" NPT	7-1/4	N/A	N/A
6	98-13/16	38-9/16	86-1/6	28-1/8	5-7/8	8-5/8	0,219	2"	6"	0,344	2" Threaded	1" NPT	1/2" NPT	9-1/8	N/A	N/A
8	115-3/4	43-1/8	97-3/8	31-11/16	7	10-3/4	0,281	2"	8"	0,344	2" Threaded	1" NPT	1/2" NPT	10-3/16	7-5/8	17-7/8
10	124	46-3/4	103-1/2	33-7/16	8	12-3/4	0,344	3"	10"	0,344	2" Threaded	1" NPT	1/2" NPT	11-13/16	8-9/16	19-11/16
12	131	55-3/16	114-3/8	34-5/8	9-5/8	16	0,344	4"	12"	0,394	2" Threaded	1" NPT	1/2" NPT	14-7/16	13-3/4	29-1/2
14	130-3/16	55-1/2	113-1/4	33-15/16	10-5/8	18	0,344	4"	14"	0,394	2" Threaded	1" NPT	1/2" NPT	15-7/16	14-7/8	31-5/8
16	104-1/16	50-1/8	86-7/8	23-9/16	11-5/8	20	0,394	4"	16"	0,5	2" Threaded	1" NPT	1/2" NPT	16-7/16	16	33-5/8
18	118-7/16	55-9/16	99-1/6	27-3/8	12-5/8	22	0,394	6"	18"	0,5	2" Threaded	1" NPT	1/2" NPT	19-1/8	17-1/8	35-7/8
20	114-5/16	48-13/16	94-11/16	28-5/8	18-7/16	24	0,5	6"	20"	0,562	4" Flanged	1" NPT	1/2" NPT	20-1/8	18-5/16	38-1/16
22	127-3/4	53-13/16	106-7/16	32-5/8	19-7/16	26	0,5	6"	22"	0,562	4" Flanged	2" NPT	1/2" NPT	21-1/8	19-15/16	41-1/16
24	135-1/16	57-5/16	112-3/16	33-7/8	21-3/4	28	0,562	8"	24"	0,562	4" Flanged	2" NPT	1/2" NPT	23-7/8	21-1/16	42-15/16
26	150	62-7/8	126-13/16	38-7/16	22-3/4	30	0,562	8"	26"	0,625	4" Flanged	2" NPT	1/2" NPT	24-7/8	22-1/4	45-3/8
28	162-5/8	65-1/8	131-1/4	39-11/16	23-3/4	32	0,562	8"	28"	0,625	4" Flanged	2" NPT	1/2" NPT	25-7/8	23-7/16	46-7/16
30	181-7/16	72-11/16	147-9/16	44-1/2	24-3/4	34	0,625	10"	30"	0,625	4" Flanged	2" NPT	1/2" NPT	27-11/16	24-5/8	48-5/8
32	186-7/8	74-15/16	162-3/16	45-3/4	25-3/4	36	0,625	10"	32"	0,625	4" Flanged	2" NPT	1/2" NPT	28-11/16	27-3/8	52-3/8
34	206-1/8	81-11/16	168-7/8	51-1/2	26-13/16	38	0,625	10"	34"	0,625	4" Flanged	2" NPT	1/2" NPT	30-1/4	29	56
36	213-13/16	85-1/16	174-3/4	52-7/8	27-13/16	40	0,625	12"	36"	0,625	4" Flanged	2" NPT	1/2" NPT	31-5/8	30-3/16	58-3/16
38	224-5/8	89-1/16	182-11/16	55-7/8	28-13/16	42	0,625	12"	38"	0,688	4" Flanged	2" NPT	1/2" NPT	32-5/8	31-5/16	60-5/16
40	235-3/16	93-1/16	194-1/2	58-7/8	29-13/16	44	0,625	12"	40"	0,688	4" Flanged	2" NPT	1/2" NPT	33-5/8	32-1/2	62-1/2
42	242	96-3/16	199-9/16	60-1/8	30-13/16	46	0,688	14"	42"	0,688	4" Flanged	2" NPT	1/2" NPT	35-3/16	33-11/16	64-11/16
44	257-13/16	101-15/16	214-5/8	64-7/8	31-13/16	48	0,688	14"	44"	0,75	4" Flanged	2" NPT	1/2" NPT	36-3/16	34-13/16	66-13/16

Sizing table values are reproduced from the user-supplied reference and have not been changed.

6. Standard Supply and Optional Additions

Standard Supply	Optional Additions
- Main barrel / reducer section - Quick opening closure or approved end closure - Pig signaler connection - Vent and drain nozzles - Support saddles / structural supports - Lifting points - Hydrostatic testing - Primer or project coating - Basic manufacturing dossier	- Kicker and bypass piping - Valves and actuators - Pressure gauges and transmitters - Closure safety interlock enhancement - Special NDT or third-party witnessing - Insulation or special coating - Skid-mounted control panel - Export preservation / long-term storage package

7. Skid-Mounted Complete Pigging Systems



Complete pigging systems can integrate the launcher or receiver with valves, bypass lines, vent/drain piping, instrumentation and structural framing on one fabricated skid.

The packaged approach reduces site fabrication, improves installation consistency and shortens commissioning time. It is suitable for permanent stations, remote installations and modular transport.

Optional provisions include local instrument panels, drain containment, electrical accessories, lifting arrangements and customized operating philosophy.

8. Quick Opening Closure

Quick Opening Closure Visuals

The following views illustrate the quick-opening closure arrangement in open and exploded form for clearer technical understanding.



Figure 3. Open quick-opening closure with internal barrel access.



Figure 4. Exploded / cutaway closure view showing component arrangement.

For frequent pigging service, a quick opening closure may be supplied with a positive locking mechanism, service-compatible seal and safety provision intended to prevent opening while residual pressure remains in the barrel.

9. Design, Inspection and Testing

Design Considerations	Inspection and Testing
Line size and schedule	Material traceability
Design pressure and temperature	WPS/PQR and welder qualification review
Service fluid and corrosion allowance	Visual and dimensional inspection
Pig type and dimensions	NDT as specified
Closure type	Hydrostatic pressure test
Nozzle schedule and orientation	Coating inspection
Support and lifting requirements	Final documentation review
Material class and coating system	Third-party witnessing when required

10. Applicable Design Basis

Depending on project scope and jurisdiction, the design basis may reference ASME B31.4, ASME B31.8, ASME Section VIII Division 1 and client specifications. The final code basis shall be confirmed in the approved datasheet and design documents.

11. Typical Applications

Pigging Operations	Industry Segments	Documentation
- Cleaning - Batching - Gauging - Dewatering / drying - Intelligent inspection	- Oil and gas - Petrochemical - Chemical - Water pipelines - Terminals and transfer lines	- Approved datasheet - GA drawing - Material certificates - NDT / hydrotest reports - Coating / packing records

12. Quality and Documentation Controls

- Each project should be manufactured against approved drawings and procedures.
- Material identification,
- weld traceability, inspection records,
- pressure-test results,
- coating records, and
- Final release documentation should be compiled into the manufacturing dossier.



**PIG LOADING &
HANDLING
EQUIPMENT**

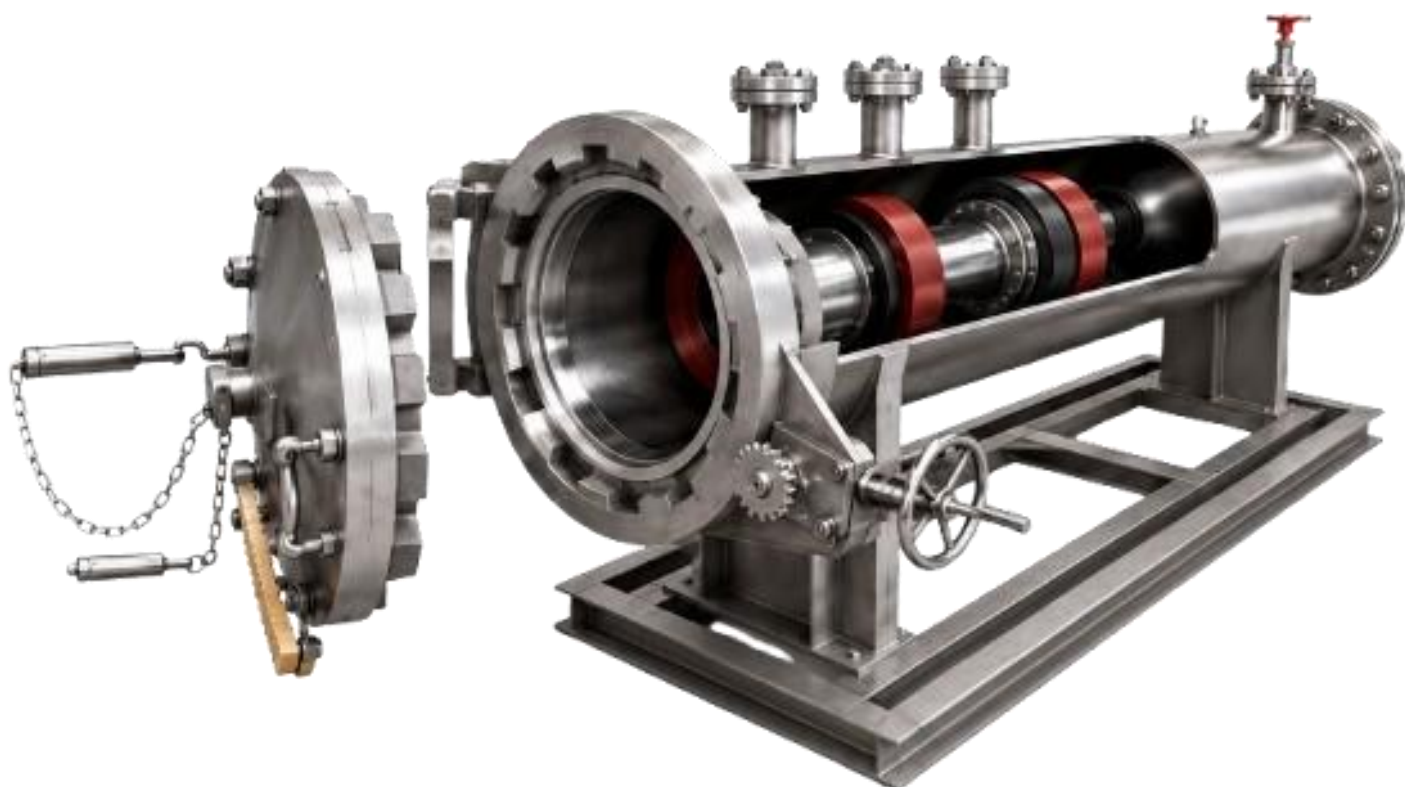


PIG LAUNCHER & RECEIVERS

PROJECT ENQUIRY GUIDE

13. Information Required with Enquiry

Required Information
Pipeline size, schedule and pressure class
Design pressure and design temperature
Operating pressure and operating temperature
Service fluid and composition
Corrosion allowance and material class
Launcher, receiver or combined-station requirement
Pig type, diameter and overall length
Closure type and opening direction
Nozzle, vent, drain, kicker and instrument requirements
Skid, valves, instruments and control-panel scope
Coating, preservation and packing requirements
Inspection, documentation and third-party witnessing requirements



LINESIG | Engineered Pipeline Pigging Solutions